

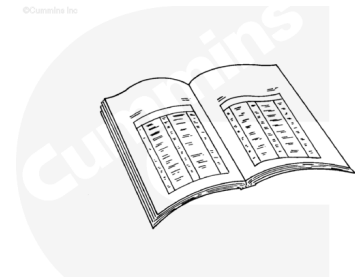
Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

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ck800wa

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

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This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid contact of hot oil with your skin.

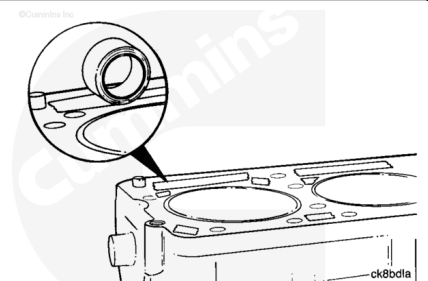
- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99->

013-009.html)

- Drain the cooling system. Refer to Procedure 008-018 in Section 18. (/qs3/pubsys2/xml/en/procedures/18/18-008-018-tr.html)
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/18/18-002-004-tr.html)
- Remove the piston cooling nozzles. Refer to Procedure 001-046 in Section 1. (/qs3/pubsys2/xml/en/procedures/18/18-001-046-tr.html)
- Drain the lubricating oil system. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/18/18-007-037-tr.html)
- Remove the oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/18/18-007-025-tr.html)
- Remove the lubricating oil pan adapter cover plate. Refer to Procedure 007-026 in Section 7. (/qs3/pubsys2/xml/en/procedures/18/18-007-026-tr.html)
- Remove the oil suction tube. Refer to Procedure 007-027 in Section 7. (/qs3/pubsys2/xml/en/procedures/18/18-007-027-tr.html)
- Remove the water bypass tube. Refer to Procedure 008-015 in Section 8. (/qs3/pubsys2/xml/en/procedures/18/18-008-015-tr.html)

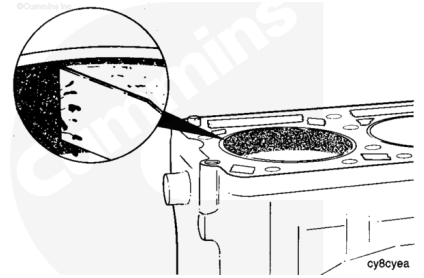
Remove

Protect the push rod galleys, coolant passages, and oil passages from contamination.



⚠ CAUTION ⚠

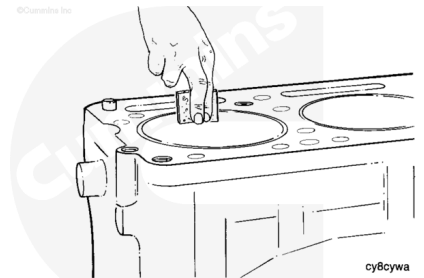
Do not use abrasive paper to remove the carbon deposits. Small particles from abrasive paper will cause severe engine damage.



Use a scraper or a similar blunt-edged tool to loosen the carbon deposits in the cylinder liner.

⚠ CAUTION ⚠

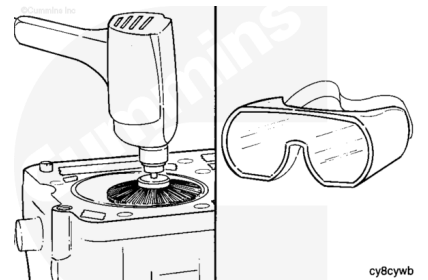
Use extreme care to make sure that no abrasive cleaners or materials are used in the piston ring travel area.



Remove the remaining carbon with a nylon abrasive pad, Scotch-Brite™ 7448, or equivalent, and solvent. The carbon **must** be removed, but the surface does **not** have to appear like new metal.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using a wire wheel to prevent serious personal injury.



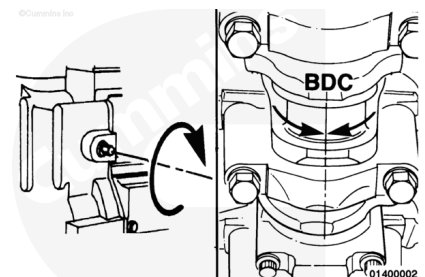
An alternative method to remove the carbon ridge is to use a high quality-steel wire wheel installed in a drill.

Do **not** use a steel wire wheel of inferior quality. The inferior wire wheel will lose steel bristles during operation and cause additional contamination.

Note : Do not use a steel wire wheel in the piston ring travel area.

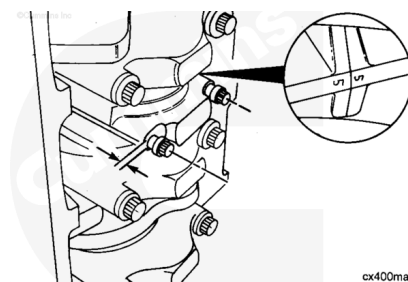
Operate the wire wheel in a circular motion to remove the deposits.

Use the barring mechanism to rotate the engine. Rotate the crankshaft to position the connecting rod at bottom dead center (BDC).



Connecting rods **must** have the cylinder number marked on both the connecting rod and the connecting rod cap on the side positioned toward the camshaft. Check the connecting rods for correct markings. Use a steel stamp and mark any rod that is **not** correctly marked.

Loosen the capscrews until there is 6 mm [$\frac{1}{4}$ inch] of clearance between the connecting rod cap and the capscrew head.

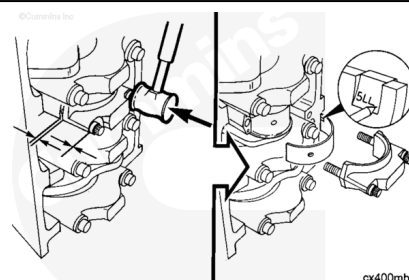


Use a mallet. Tap the connecting rod capscrews until the connecting rod cap and connecting rod separate. Remove the capscrews and the connecting rod cap.

Remove the lower rod bearing.

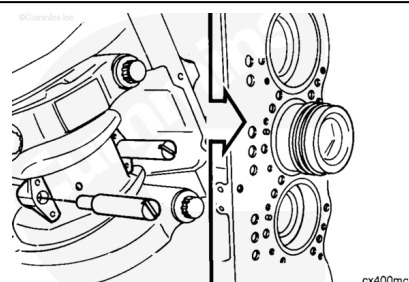
Use an awl to mark the bearing position in the tang area.

Mark the bearing for future identification or for possible failure analysis.



Install two connecting rod guide pins, Part Number 3375098, or equivalent, into the connecting rod.

Push the piston and connecting rod up until the piston rings are above the cylinder liner.



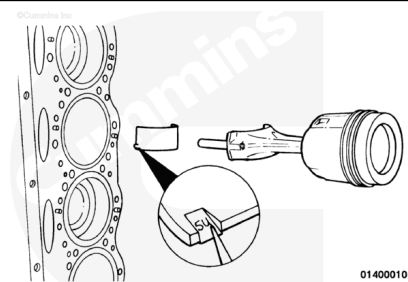
⚠ CAUTION ⚠

Place the piston and rod assembly in a rack to prevent damage to the piston and rod assembly.

Remove the piston and rod assembly.

Remove the upper connecting rod bearing.

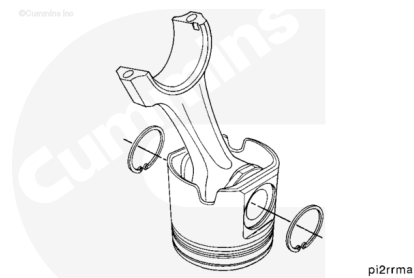
Use an awl to mark the bearing position in the tang area.



Disassemble

Remove the piston rings. Refer to Procedure 001-047 in Section 1. (</qs3/pubsys2/xml/en/procedures/18/18-001-047.html>)

Remove the snap rings from both sides of the piston with internal snap ring pliers.

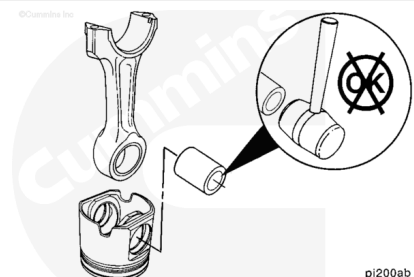


⚠ CAUTION ⚠

Do not use a hammer to remove the piston pin. The piston can distort, causing it to seize in the cylinder liner.

⚠ CAUTION ⚠

Do not drop the piston while removing the piston pin. Personal injury and damage to the piston can result.



Remove the piston pin and separate the connecting rod and piston.

Clean and inspect the piston. Refer to Procedure 001-043 in Section 1. (</qs3/pubsys2/xml/en/procedures/18/18-001-043.html>)

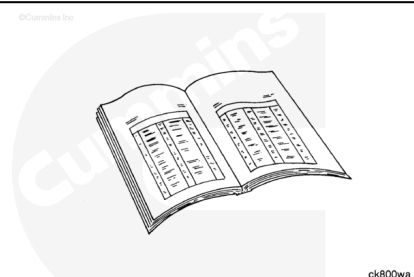
Clean and Inspect for Reuse

Clean and inspect the connecting rods for reuse. Refer to Procedure 001-014 in Section 1.

(</qs3/pubsys2/xml/en/procedures/18/18-001-014.html>)

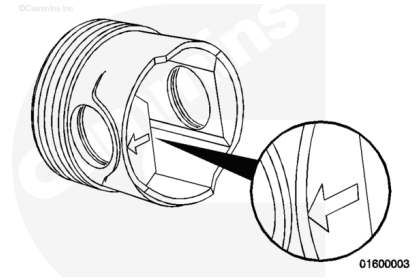
Clean and inspect the pistons for reuse. Refer to Procedure 001-043 in Section 1.

(</qs3/pubsys2/xml/en/procedures/18/18-001-043.html>)



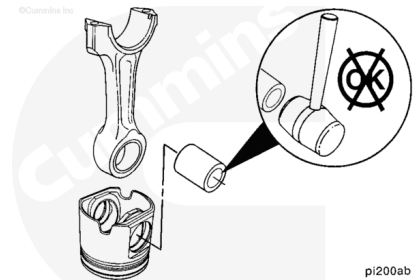
Assemble

There are two different types of pistons for this engine. Offset pin pistons can be identified by the cast arrow on the bottom of the pin bore.



⚠ CAUTION ⚠

Do not use a hammer to install the piston pin. The piston can distort, causing it to seize in the cylinder liner.



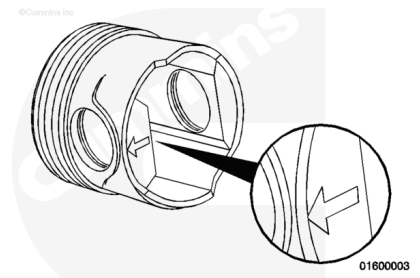
Install a new snap ring into one of the piston pin bores.

The snap ring **must** be seated completely in the piston groove to prevent engine damage during engine operation.

Lubricate the piston pin and connecting rod bushing with clean engine oil.

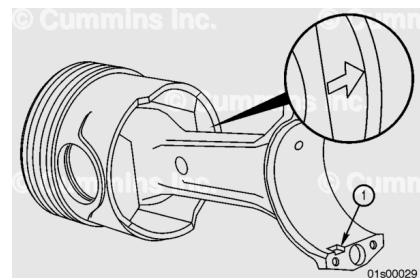
Align the pin bore of the rod with the pin bore of the piston skirt and crown, and install the piston pin.

Offset pin pistons have a cast arrow on the bottom of the pin bore.



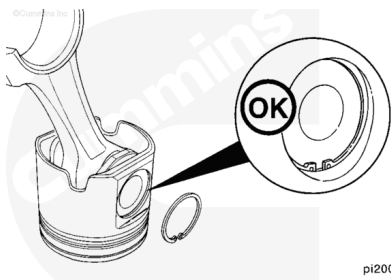
Assemble the offset pin with the arrow pointing toward the front of the engine.

Note : When properly assembled, the arrow is on the opposite side of the lock tang (1) on the connecting rod.

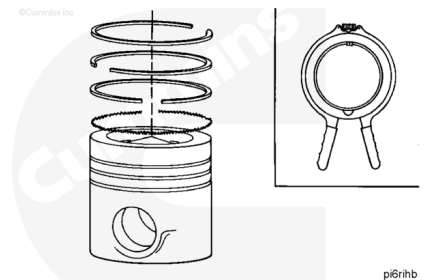


Install a new snap ring in the piston pin bore.

The snap ring **must** be seated completely in the piston groove to prevent engine damage during operation.



Install the piston rings. Refer to Procedure 001-047 in Section 1. (</qs3/pubsys2/xml/en/procedures/18/18-001-047.html>)



Install

⚠ CAUTION ⚠

To reduce the possibility of engine damage, do not lubricate the back side of the bearing shells.

Clean the connecting rod and the bearing shells with a lint free cloth.

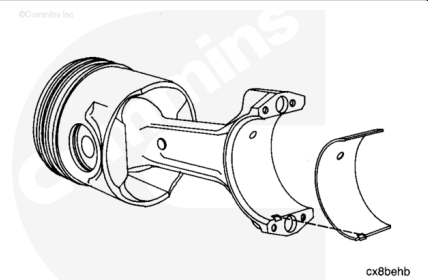
Note : All of the connecting rod bearings are identical.

Note : Do not reuse connecting rod bearings if either the connecting rod or crankshaft has been changed.

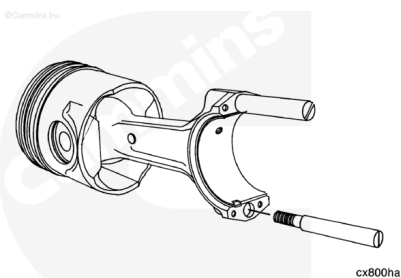
Install the connecting rod bearing. Be sure the tang is positioned as shown. The end of the bearing **must** be even with the cap mounting surface.

Lubricate the connecting rod bearing surface with clean engine oil.

The connecting rod bearings **must** be installed in their original locations if new connecting rod bearings are **not** used.

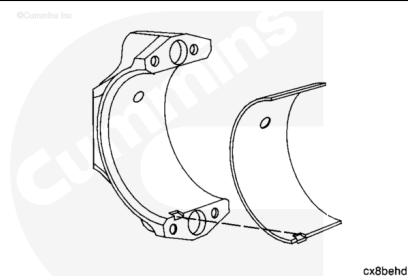


Install two connecting rod guide pins, Part Number 3375098, in the connecting rod. The guide pins will aid the assembly process and protect the crankshaft.



⚠ CAUTION ⚠

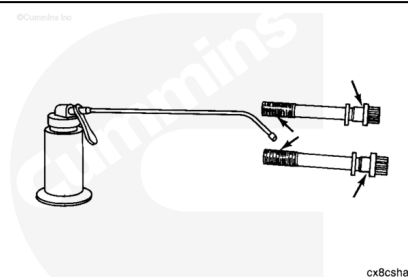
The connecting rods and connecting rod caps are not interchangeable. The connecting rods and the connecting rod caps are machined as an assembly. Damage to the engine will result if they are mixed.



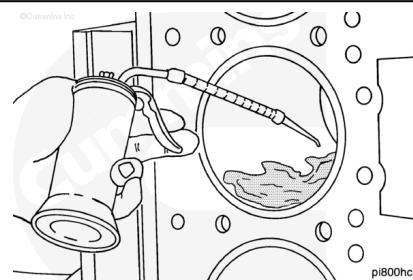
Install the lower bearing shell in the connecting rod cap. Be sure the tang of the bearing shell is in the slot of the connecting rod cap and the end of the bearing is even with the connecting rod cap surface.

Use clean lubricating oil to lubricate the connecting rod capscrews and washers as shown.

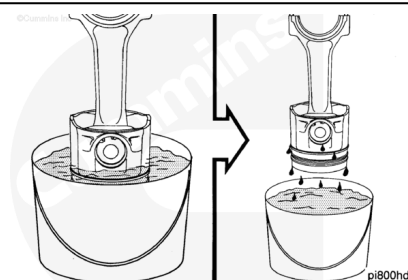
Install the washers and capscrews in the connecting rod caps.



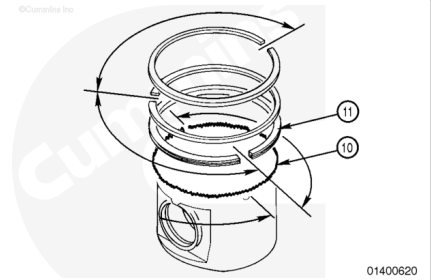
Lubricate the cylinder liner with clean engine oil. The entire bore **must** be lubricated.



Immerse the piston in engine oil until the rings are covered. Allow the excess oil to drip off the assembly.



Make sure the piston ring gap position is still correct.

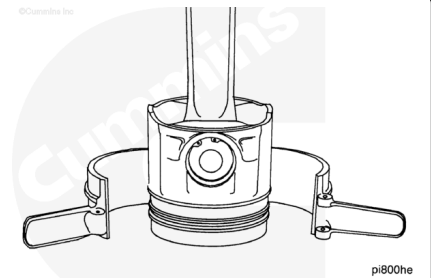


⚠ CAUTION ⚠

To reduce the possibility of engine damage, make sure the piston rings fit correctly in the piston.

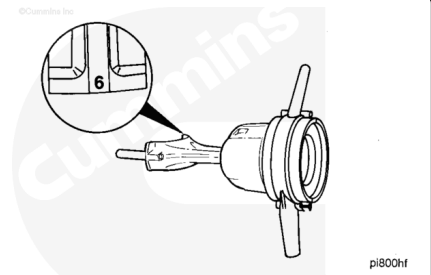
Use a piston ring compressor, Part Number 3375342, or equivalent. Install the ring compressor on the piston.

Note : The ring compressor has a tapered bore. The small end of the taper must be positioned toward the piston skirt.



⚠ CAUTION ⚠

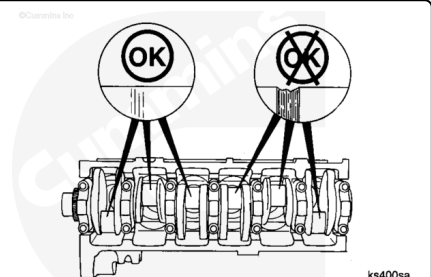
To reduce the possibility of engine damage, the cylinder number on the connecting rod and connecting rod cap must be the same and the side of the connecting rod with the cylinder number (bearing tang side) must be toward the camshaft.



Rotate the crankshaft until the journal for the rod being installed is at bottom dead center (BDC).

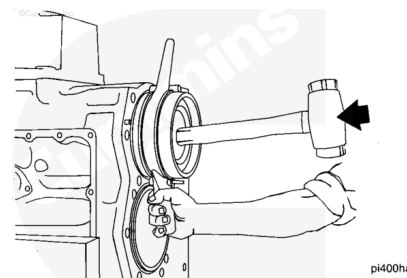
Note : If the engine has a crankshaft with bolt-on counterweights, the journal must be at top dead center (TDC).

Check the crankshaft rod journals for damage.



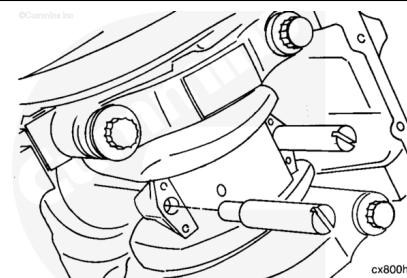
Install the connecting rod and piston until the ring compressor touches the block. Align the rod with the crankshaft journal.

Hold the ring compressor firmly against the block. Use a wooden hammer handle to push the piston into the liner.



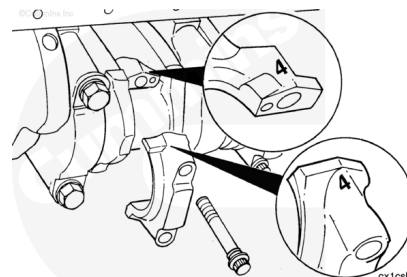
Push the piston into the bore until the rod bearing contacts the crankshaft journal.

Remove the guide pins.



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To reduce the possibility of engine damage, the cylinder number on the connecting rod and connecting rod cap must be the same and the side of the connecting rod with the cylinder number (bearing tang side) must be toward the camshaft.



Install the connecting rod cap.

Tighten the capscrews alternately and evenly to pull the cap over the dowel pins. Use the following steps to tighten the capscrews.

Torque Value:

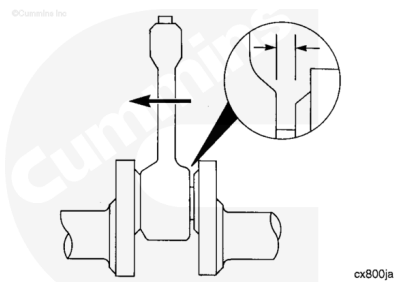
1. 102 n•m [75 ft-lb]
2. 197 n•m [145 ft-lb]
3. 292 n•m [215 ft-lb]
4. 366 n•m [270 ft-lb]
5. Loosen both capscrews to remove all tension.
6. 102 n•m [75 ft-lb]
7. 149 n•m [110 ft-lb]

Plus 90° to achieve correct bolt stretch.

Check the side clearance between the rod and the crankshaft.
The rod **must** move freely from side-to-side.

Measurements over 0.51 mm [0.020 in] **must** be measured with a dial indicator.

If the side clearance is greater than 0.51 mm [0.020 in], do **not** reuse the parts, unless the same connecting rod is used on the same crankshaft journal in the same engine. If the parts are reused on the same crankshaft connecting rod journal in the same engine, the maximum allowable side clearance, measured with a dial indicator, is 5.08 mm [0.200 in].



Connecting Rod and Crankshaft Side Clearance, New or Remanufactured Parts

mm		in
0.20	MIN	0.008
0.36	MAX	0.014

Connecting Rod and Crankshaft Clearance, Used Parts

mm		in
0.51	MAX	0.020

Connecting Rod and Crankshaft Clearance, Same Connecting Rod and Crankshaft Reused on Same Journal of Crankshaft

mm		in
5.08	MAX	0.200

Finishing Steps

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- Install the oil suction tube. Refer to Procedure 007-027 in Section 7. (</qs3/pubsys2/xml/en/procedures/18/18-007-027-tr.html>)
- Install the oil pan adapter cover plate. Refer to Procedure 007-026 in Section 7. (</qs3/pubsys2/xml/en/procedures/18/18-007-026-tr.html>)
- Install the oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/18/18-007-025-tr.html>)
- Install the cylinder head. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/18/18-002-004-tr.html>)
- Install the piston cooling nozzles. Refer to Procedure 001-046 in Section 1. (</qs3/pubsys2/xml/en/procedures/18/18-001-046-tr.html>)
- Install the water bypass tube. Refer to Procedure 008-013 in Section 8. (</qs3/pubsys2/xml/en/procedures/18/18-008-013.html>)
- Fill the engine with clean oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/18/18-007-037-tr.html>)
- Fill the cooling system. Refer to Procedure 008-018 in Section 18. (</qs3/pubsys2/xml/en/procedures/18/18-008-018-tr.html>)
- Connect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Operate the engine to 70°C [160°F] coolant temperature and check for leaks.

